

Pituitary and Hypothalamus

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Learning Objectives

1. Describe the major structures of the hypothalamic-pituitary axis.
2. Describe the hormones produced in the anterior and posterior pituitary and targets of each.
3. Describe the production, secretion, regulation, and functions of growth hormone and IGF-1 within the GH/IGF-1 axis, including the insulin-like parallels of IGF-1 .
4. Identify the insulin-like functions and the anti-insulin functions of growth hormone.

Learning Objectives

5. Define dwarfism, gigantism, and acromegaly.
6. Identify the signs and symptoms of acromegaly.

Pituitary

3 glands connected to the hypothalamus
by the pituitary stalk

anterior:	endocrine
intermediate:	vestigial (humans)
posterior:	neuroendocrine

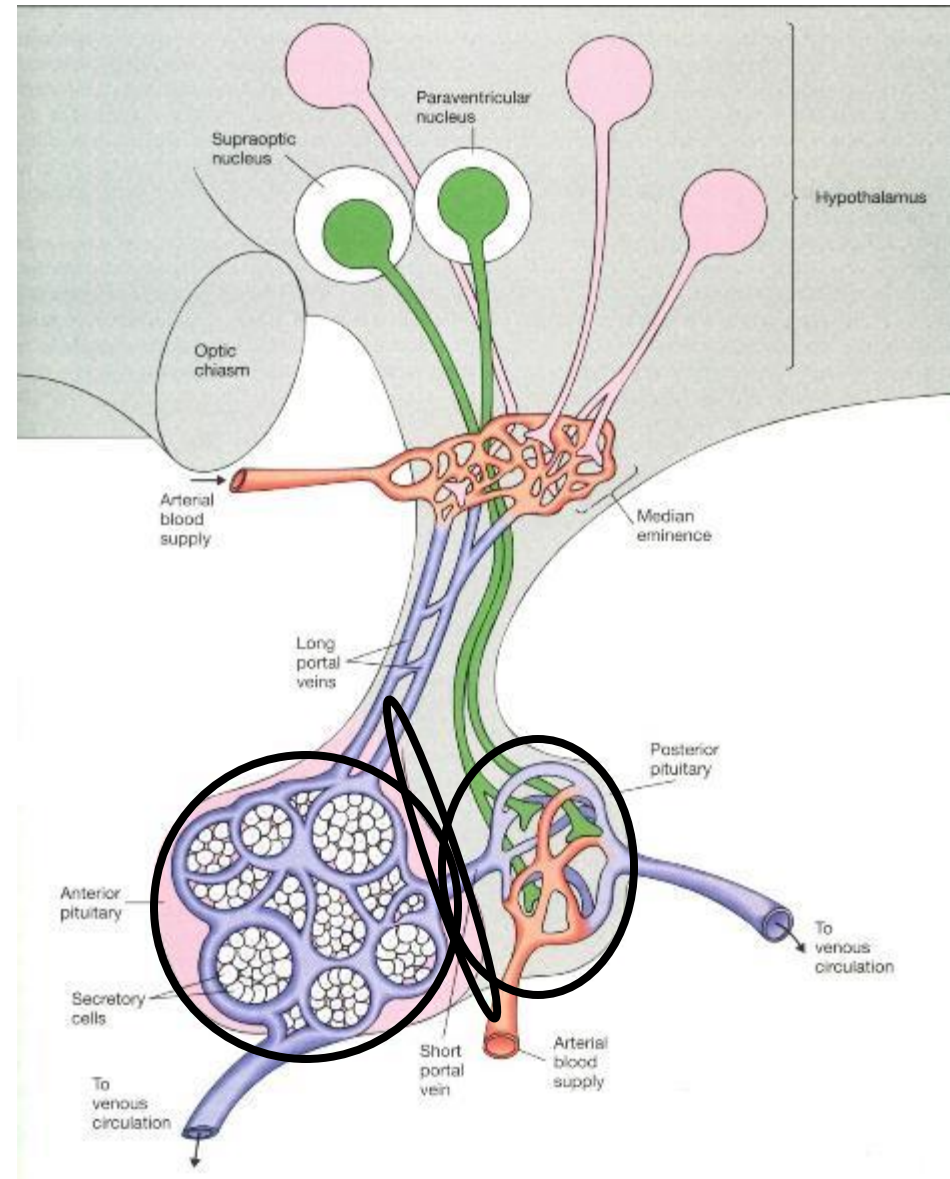


TABLE 1

Cells in Anterior pituitary lobe	Hormone secreted	Regulated by ↑ stimulation ↓ inhibition	Hypothalamic hormone
corticotropes	adrenocorticotropin (ACTH)	↑	corticotropin-releasing hormone (CRH)
thyrotropes	thyroid-stimulating hormone (TSH)	↑ ↓	thyrotropin releasing hormone (TRH) somatostatin
gonadotropes	luteinizing hormone (LH) follicle-stimulating hormone (FSH)	↑	gonadotropin-releasing hormone (GnRH)
somatotropes	growth hormone (GH)	↑ ↓	growth hormone-releasing hormone (GHRH) somatostatin
lactotropes	prolactin (PRL)	↑ ↓ ↓	prolactin-releasing factor (PRF) prolactin-inhibitory hormone (PIH) somatostatin
Neuron bodies in hypothalamus	Synthesize		Released in posterior pituitary lobe
supraoptic nuclei	antidiuretic hormone (ADH, vasopressin)		by action potentials
paraventricular nuclei	oxytocin		by action potentials

Anterior Pituitary

anterior pituitary: no direct innervation

certain hypothalamic neuron terminals end at the median eminence, where arterial blood feeds into anterior pituitary

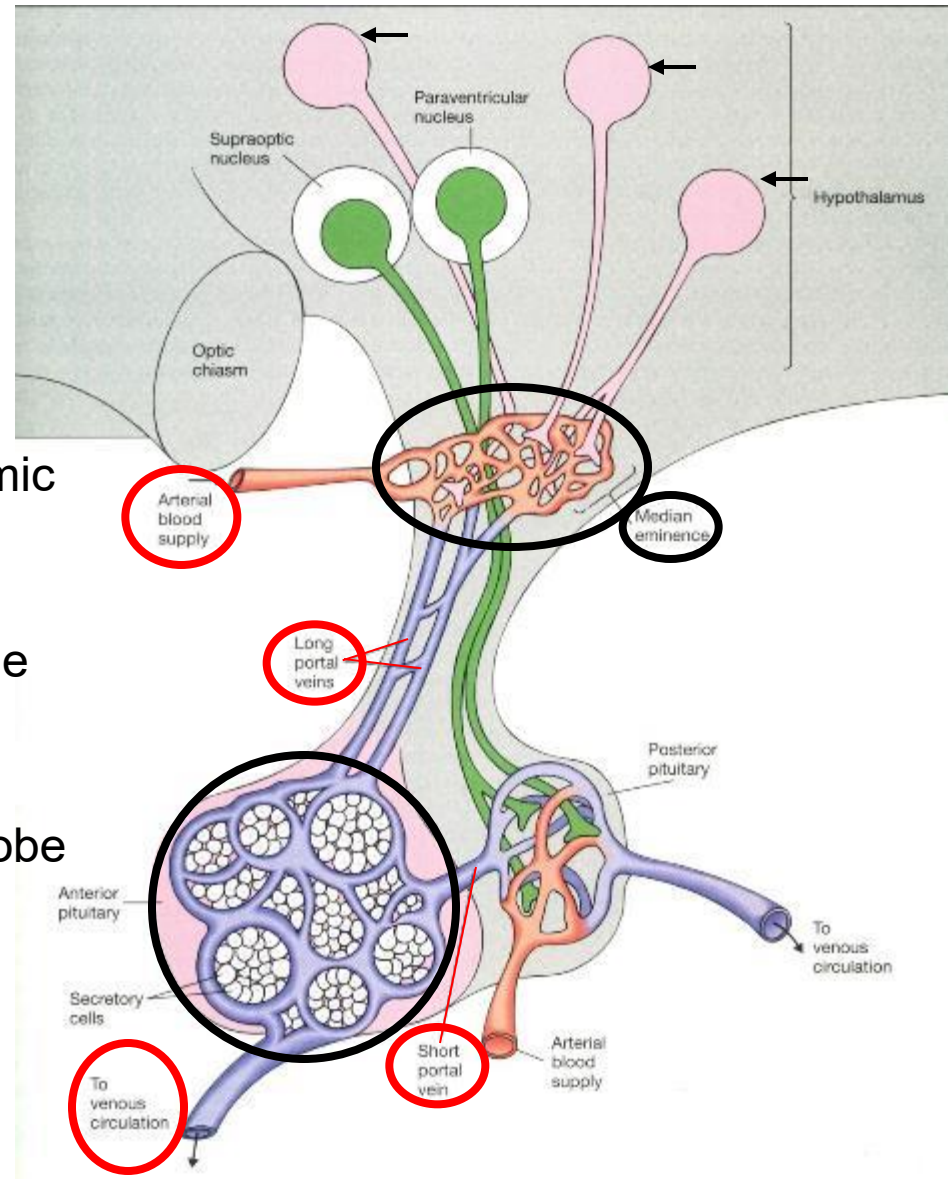
hormones synthesized in these hypothalamic neurons are:

- carried from the median eminence through the long portal veins in the pituitary stalk

- delivered to the capillary bed that perfuses the anterior pituitary lobe

anterior pituitary capillary bed feeds into the venous circulation

short portal system - venous connection between posterior and anterior pituitary



Anterior Pituitary

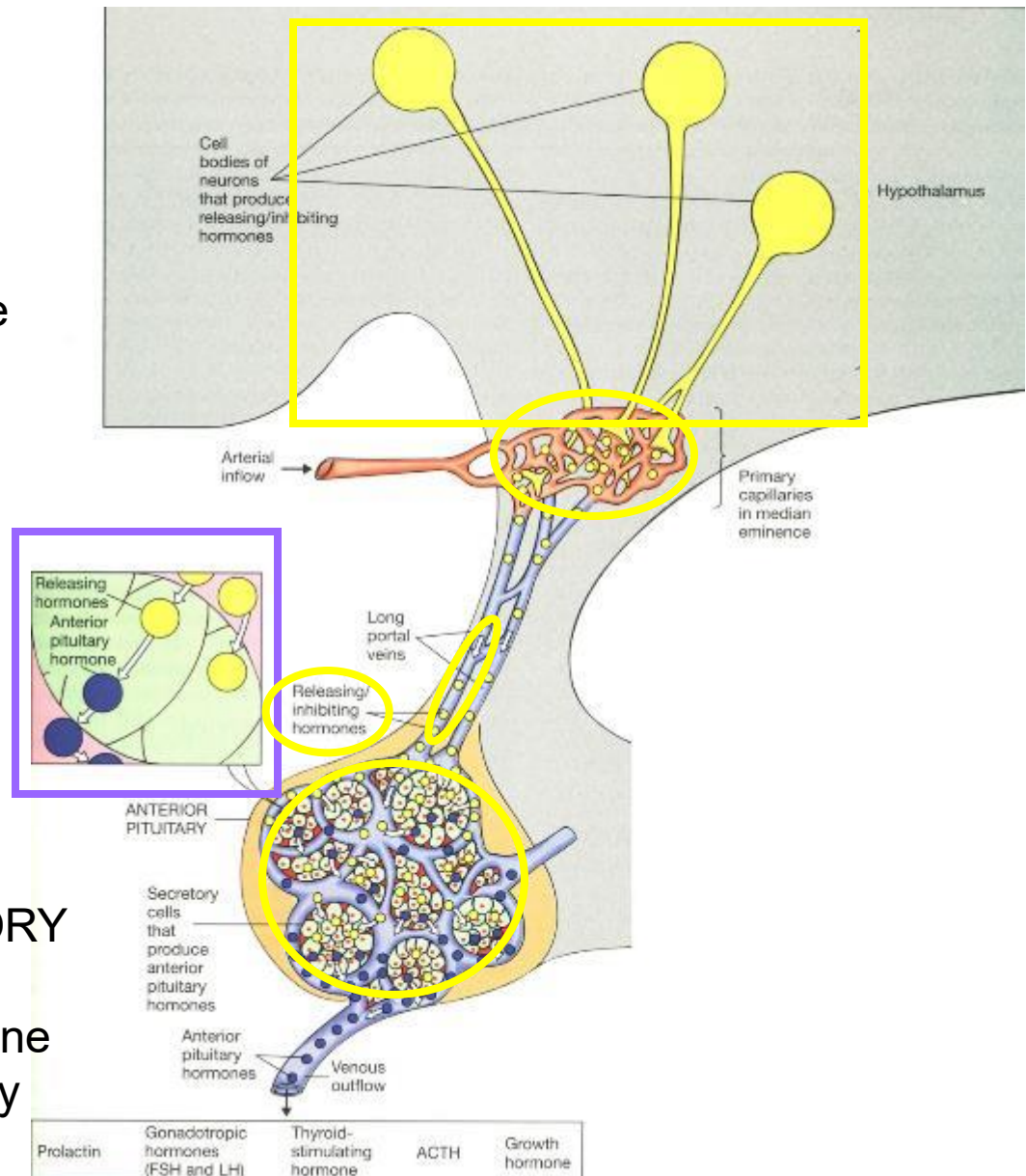
regulatory proteins synthesized in hypothalamic neurons:

released at median eminence
carried by long portal veins
perfuse the anterior pituitary

regulatory proteins act on endocrine cells within the anterior pituitary to stimulate or inhibit hormone secretion

Specificity of this response?

RATIO of STIMULATORY:INHIBITORY hypothalamic hormones determines concentration of hormone secreted from the anterior pituitary



Posterior Pituitary

neuroendocrine hormone release

supraoptic nucleus synthesizes ADH

paraventricular nucleus synthesizes oxytocin

ADH and oxytocin:

- transported down axons

- released by action potentials at the posterior pituitary capillary bed

- enter the venous circulation

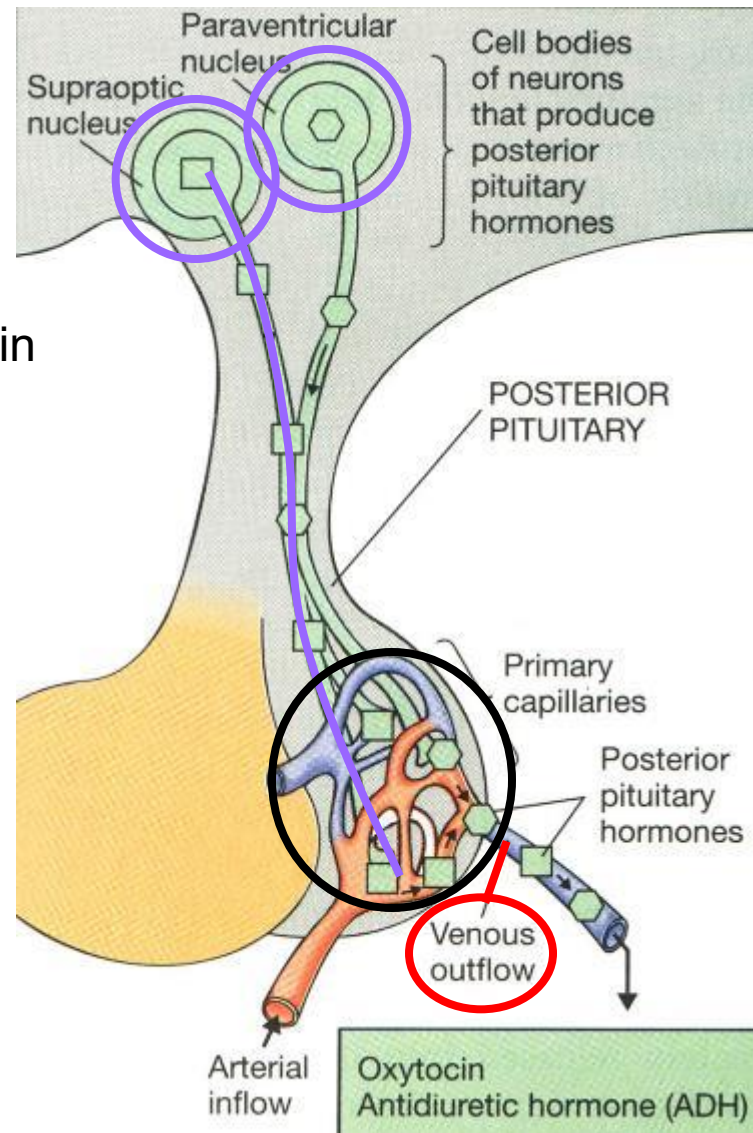


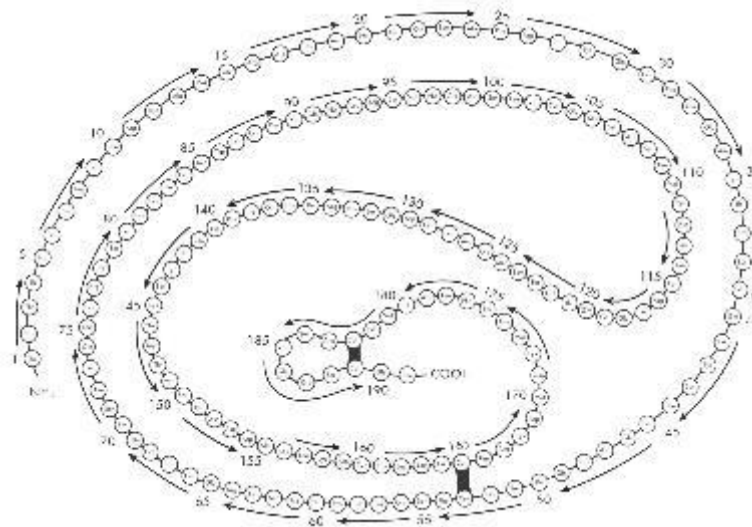
TABLE 1

Cells in Anterior pituitary lobe	Hormone secreted	Regulated by Hypothalamic hormone ↑ stimulation ↓ inhibition
20%	corticotropes	adrenocorticotropin (ACTH)
5%	thyrotropes	thyroid-stimulating hormone (TSH)
	gonadotropes	luteinizing hormone (LH) follicle-stimulating hormone (FSH)
50%	somatotropes	growth hormone (GH)
	lactotropes	prolactin (PRL)
Neuron bodies in hypothalamus	Synthesize	Released in posterior pituitary lobe
supraoptic nuclei	antidiuretic hormone (ADH, vasopressin)	by action potentials
paraventricular nuclei	oxytocin	by action potentials

Growth Hormone

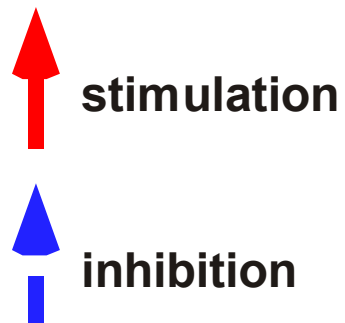
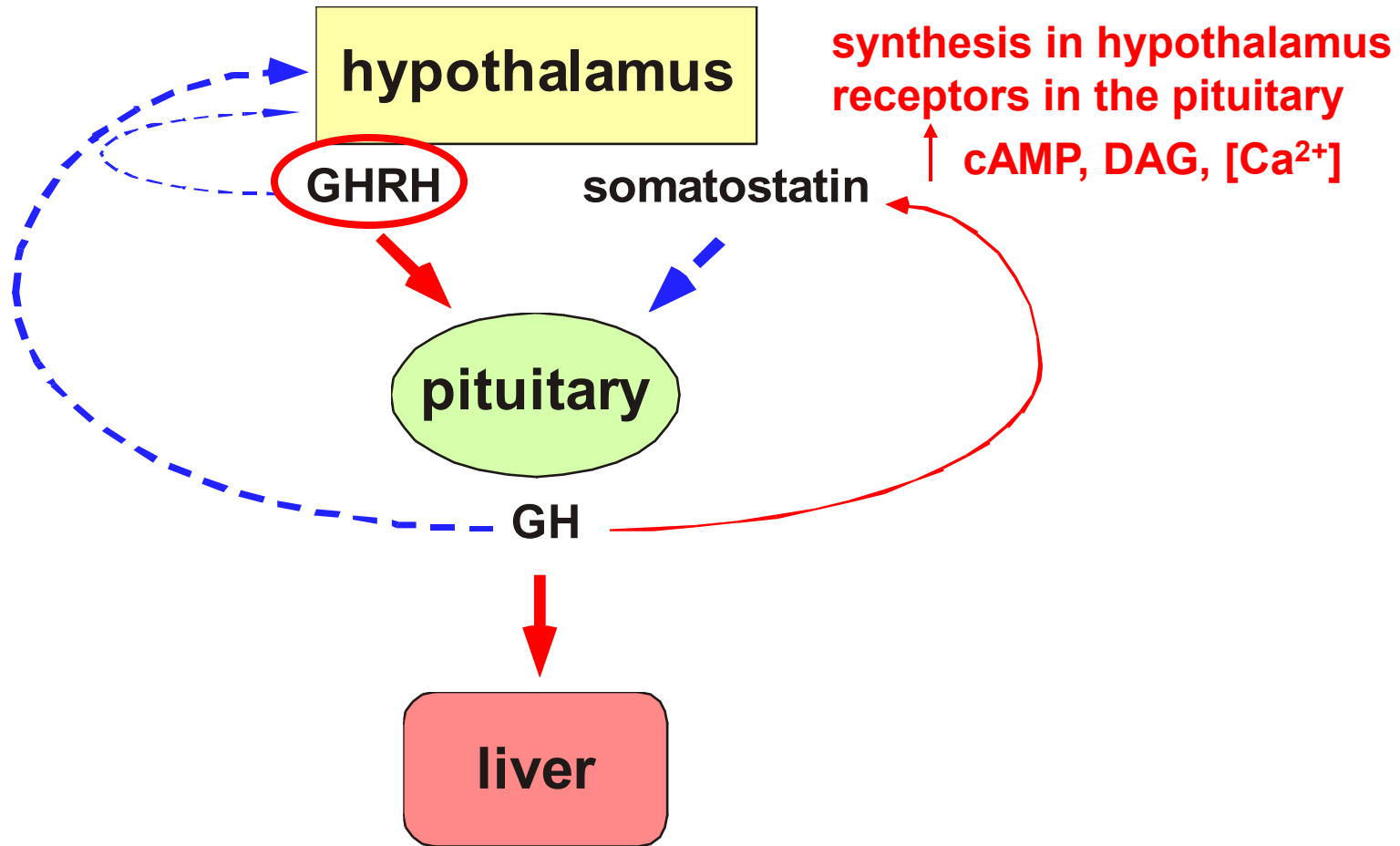
regulates: longitudinal growth of bones
glucose, amino acid, fatty acid metabolism

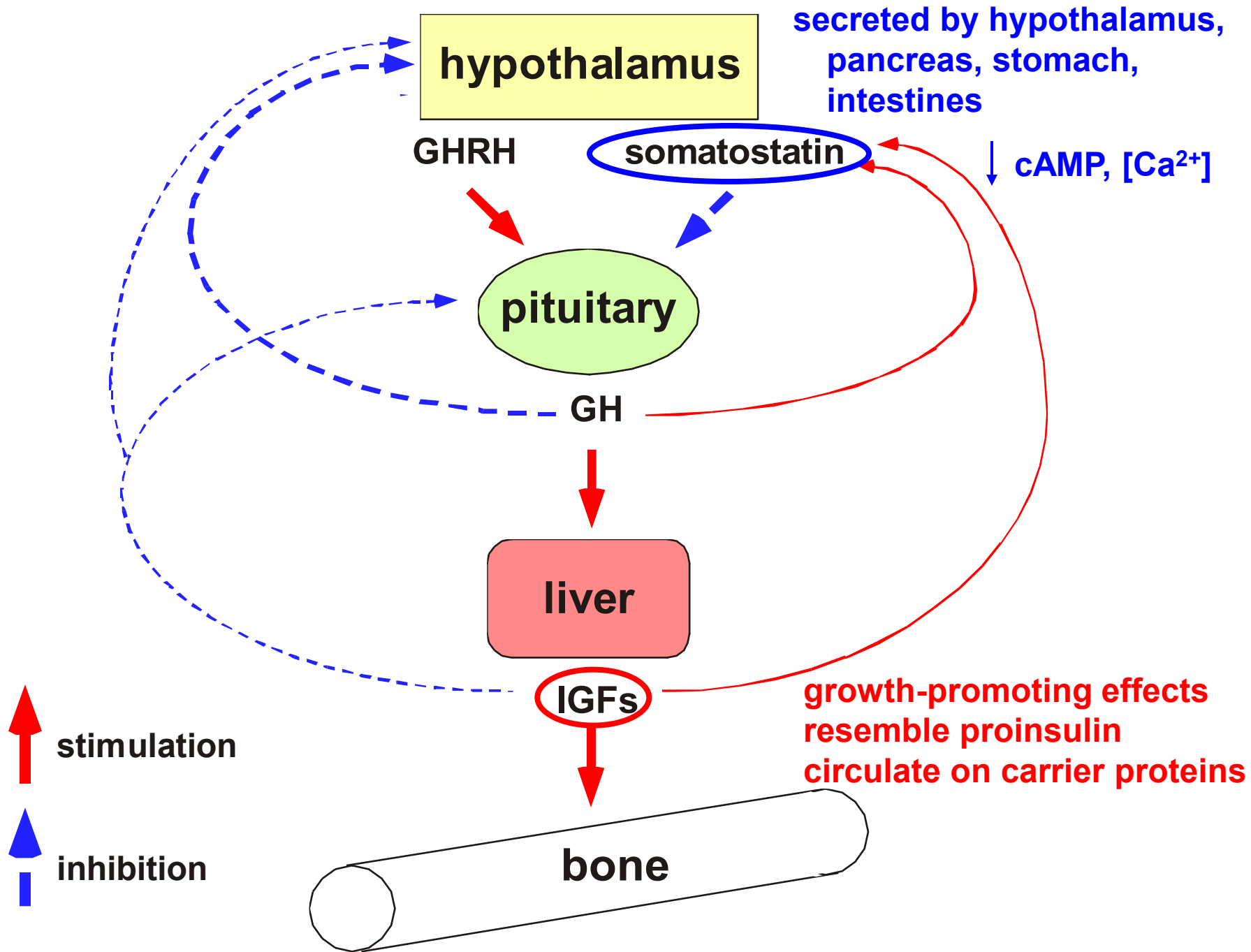
synthesis: anterior pituitary
prehormone - signal sequence removed before
storage in granules
single chain protein with 2 disulfide bonds

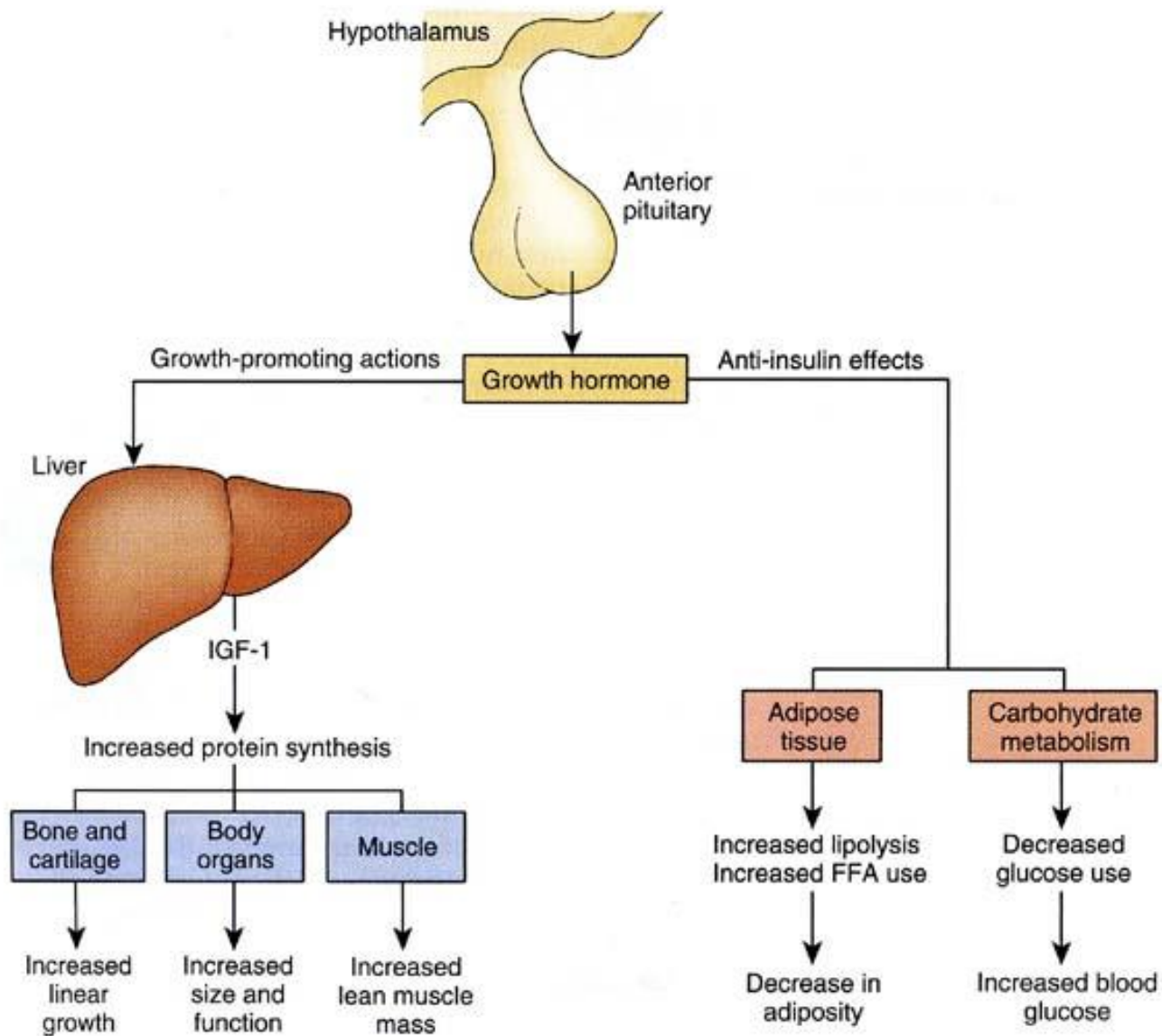


Growth Hormone

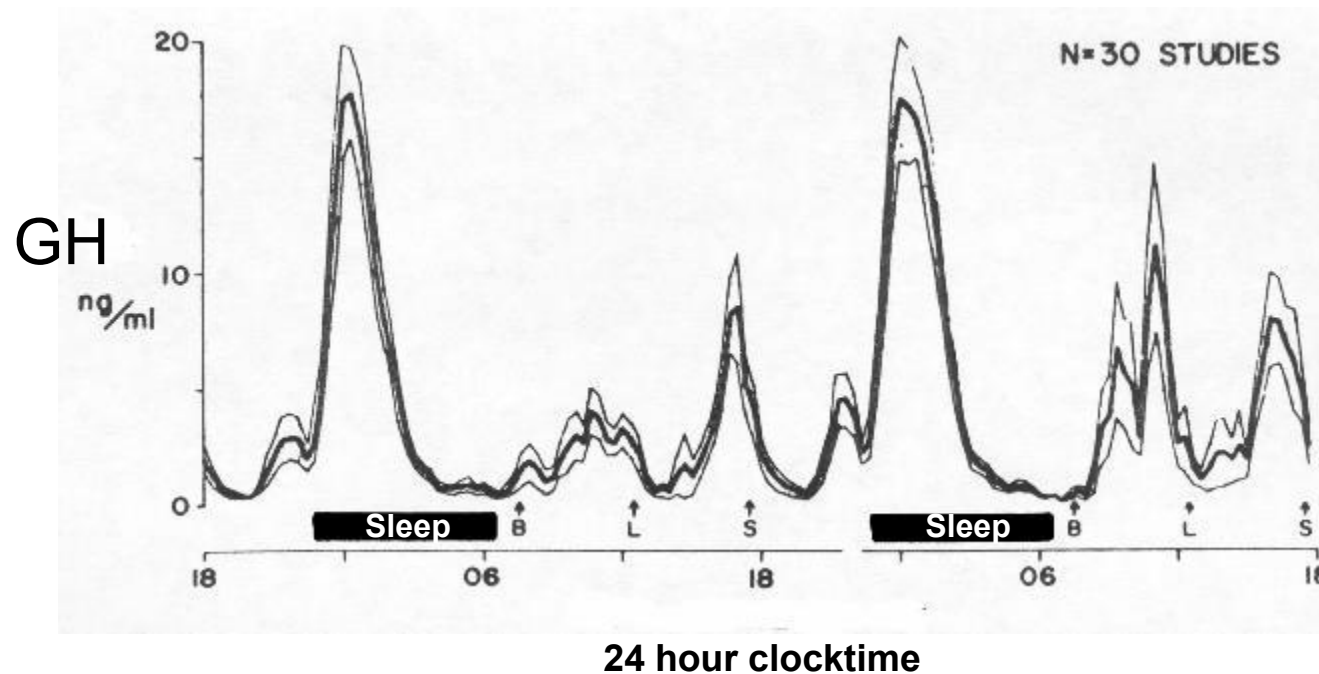
regulates:	longitudinal growth of bones glucose, amino acid, fatty acid metabolism
synthesis:	anterior pituitary prehormone - signal sequence removed before storage in granules single chain protein with 2 disulfide bonds
secretion:	pulsatile <1 - 50 ng/ml 70% secreted during slow wave sleep
circulating $t_{1/2}$:	6-20 min metabolized by liver



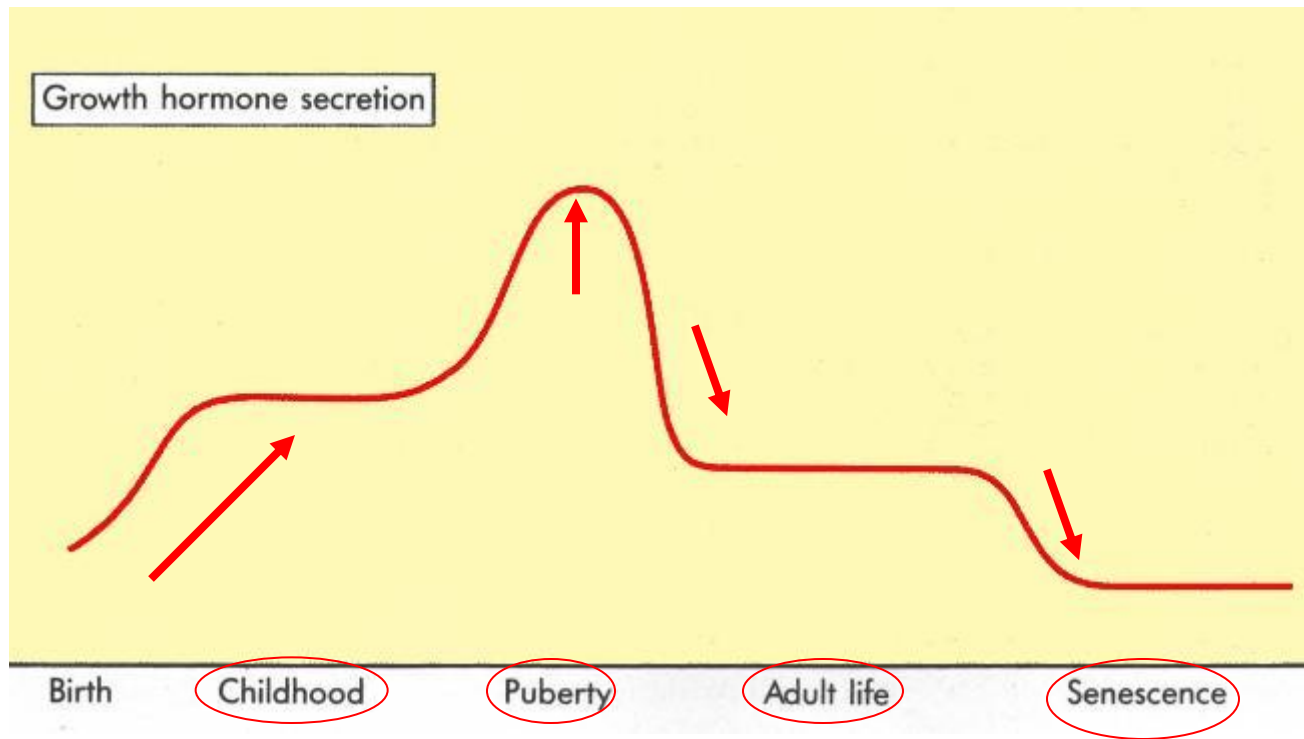




**From Porth,
2005**



pulsed release of GHRH alters GHRH:somatostatin ratio
periodic absence of GH necessary to minimize insulin antagonistic actions
(enhanced lipolysis, impaired glucose uptake and utilization)
oscillations: peaks during slow wave sleep
naps elicit GH secretion
deprivation of slow wave sleep attenuates GH secretion



women are more responsive (estrogen effect)
tall individuals more responsive than average height individuals

Mediators of Growth Hormone Secretion

stimulators

hypoglycemia
slow wave sleep
amino acids (Arg)
exercise
stress
puberty

inhibitors

glucose
REM sleep
somatostatin
growth hormone
insulin-like growth factors

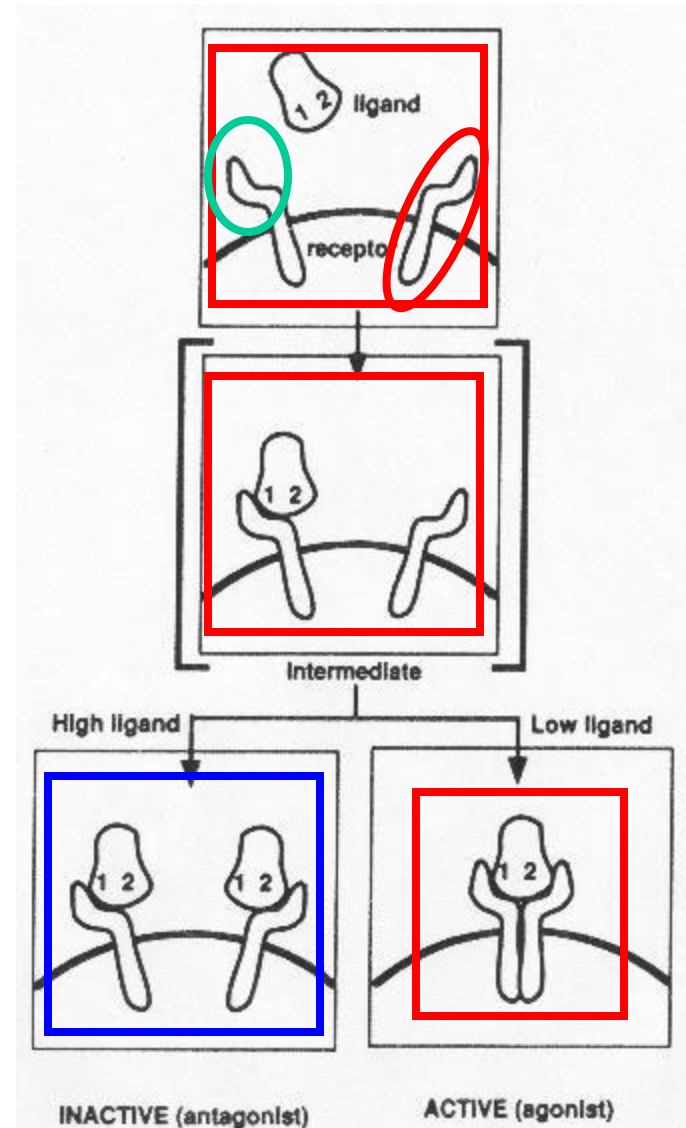
GH Receptor

found in most tissues of body

single chain protein receptor
dimerizes by binding different domains of GH
excess ligand inhibits dimerization

synthesis stimulated by low [GH]
also by insulin, estrogen
synthesis inhibited by high [GH]

GH binding protein - extracellular domain
of GHR
binds ~30% GH in reversible fashion

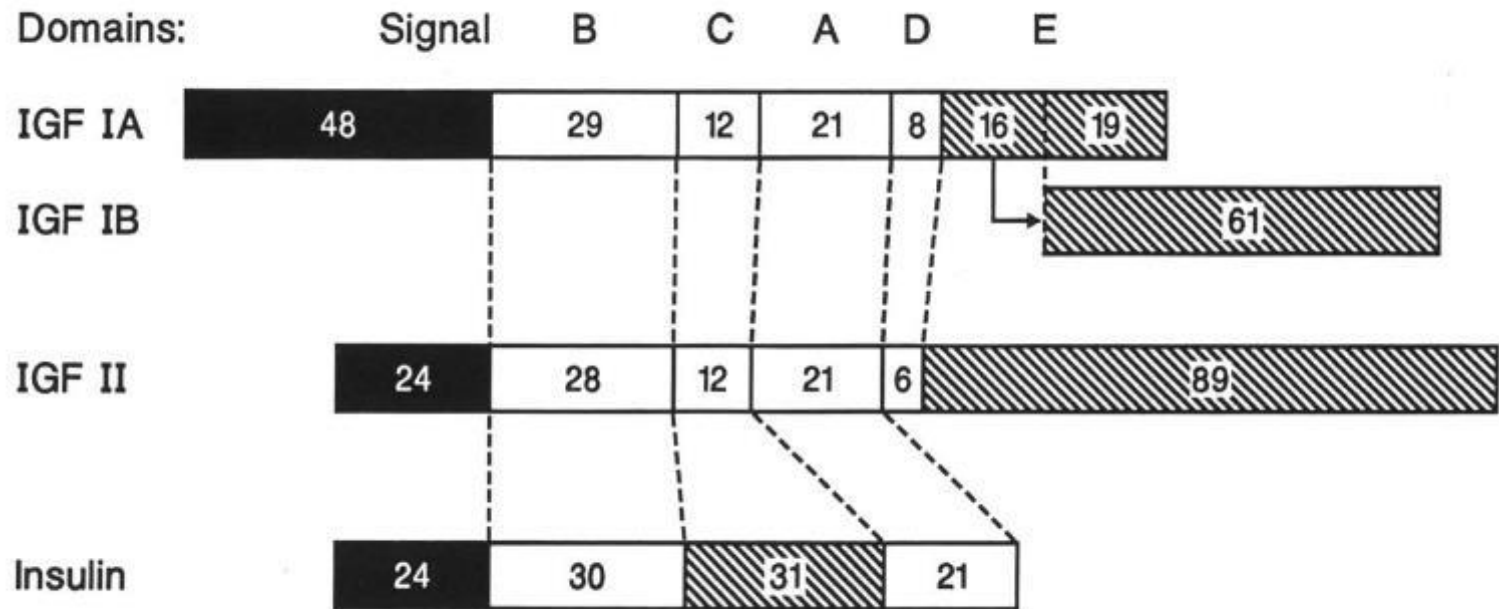


Insulin-like Growth Factors (somatomedins)

synthesized in the liver

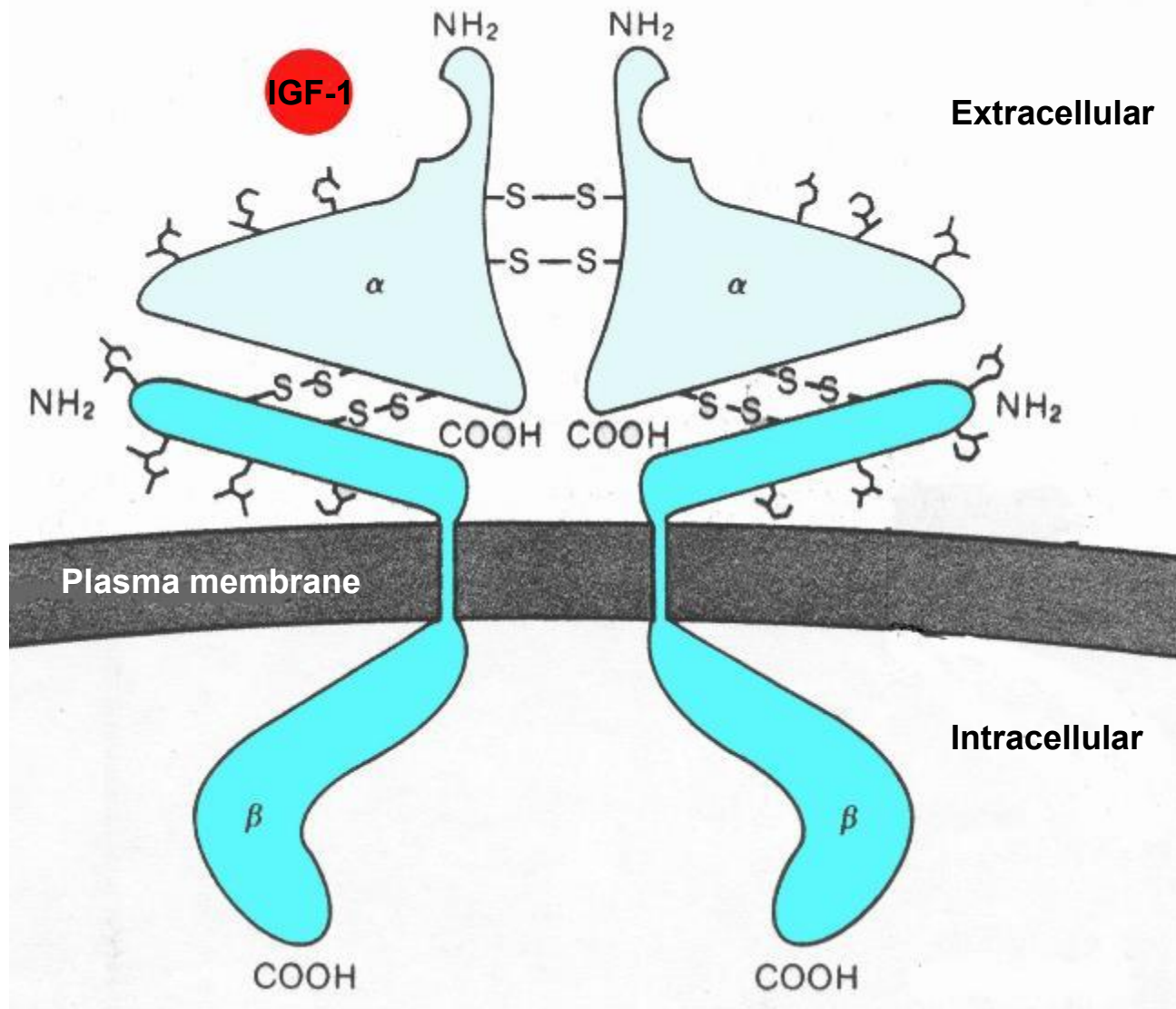
resemble proinsulin

responsible for most GH growth-promoting activity that occurs 12-18 hr
after GH increase



IGF-I Receptor

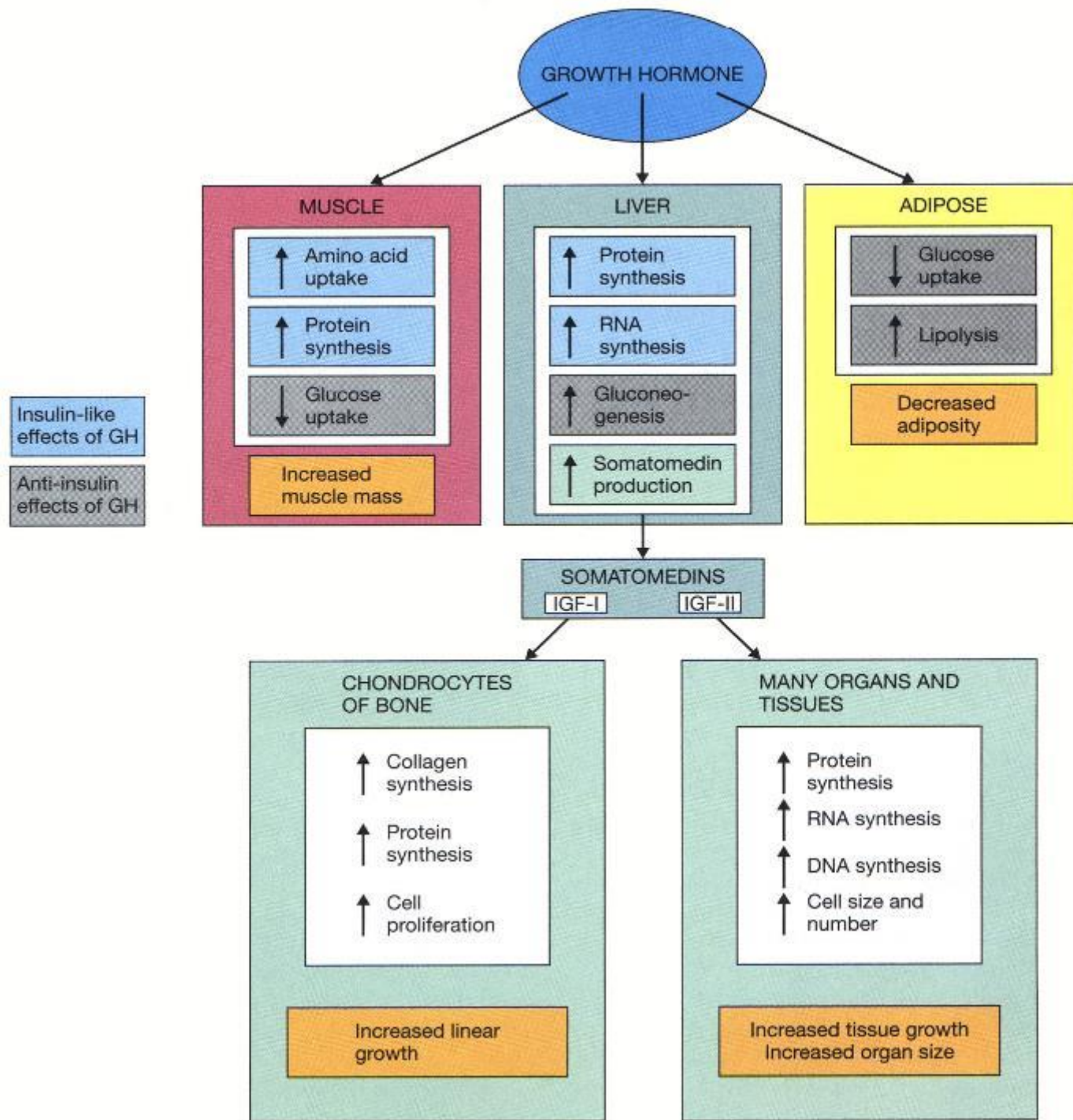
looks like insulin receptor



IGF binding proteins (IGFBPs) bind ~75% IGFs

protect IGFs from degradation - potentiate IGF activity

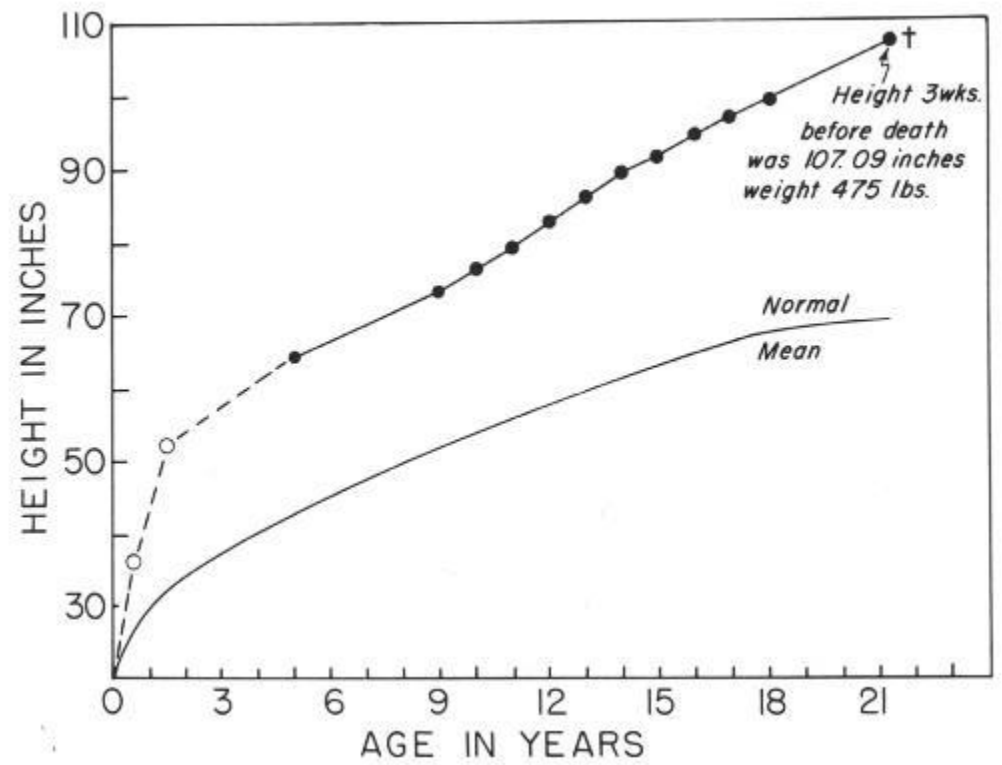
$t_{1/2}$ much longer than GH



GH Pathophysiology

Gigantism/Acromegaly

excess GH secretion: onset during childhood, puberty - **Gigantism**



GH Pathophysiology

Gigantism/Acromegaly

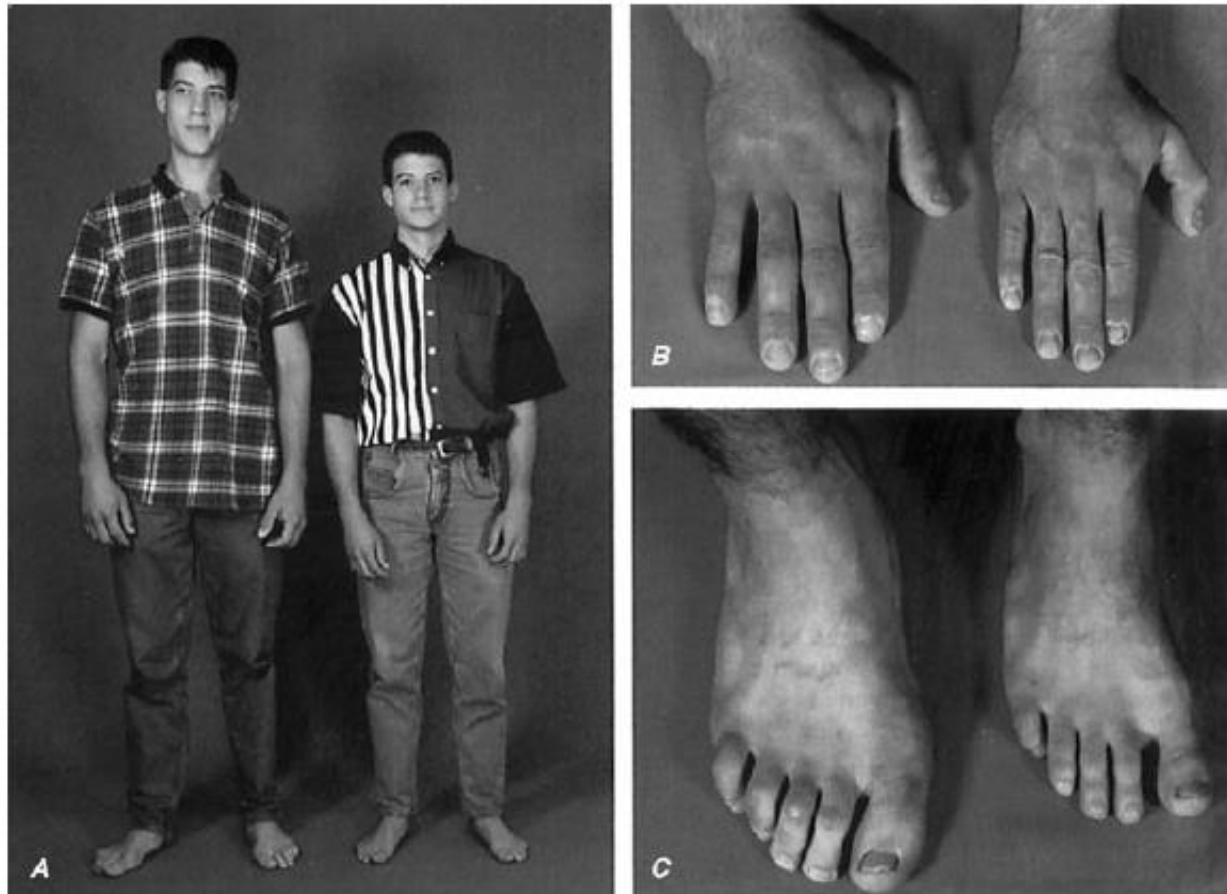
excess GH secretion: onset during childhood, puberty - **Gigantism**
onset after puberty - **Acromegaly**



widened fingers, toes, hands, feet
 bony ridge above eyes
 prominent lower jaw
 coarse facial features
 enlarged, muscular tongue
 thick skin
 sparse subcutaneous fat
 enlarged organs (heart)
 accelerated atherosclerosis



Figure 318-8



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Features of acromegaly/gigantism. A 22-year-old man with gigantism due to excess growth hormone is shown to the left of his identical twin. The increased height and prognathism (A) and enlarged hand (B) and foot (C) of the affected twin are apparent. Their clinical features began to diverge at the age of approximately 13 years.

(Reproduced from R Gagel, IE McCutcheon: *N Engl J Med* 324:524, 1999, with permission.)

GH Pathophysiology

Gigantism/Acromegaly

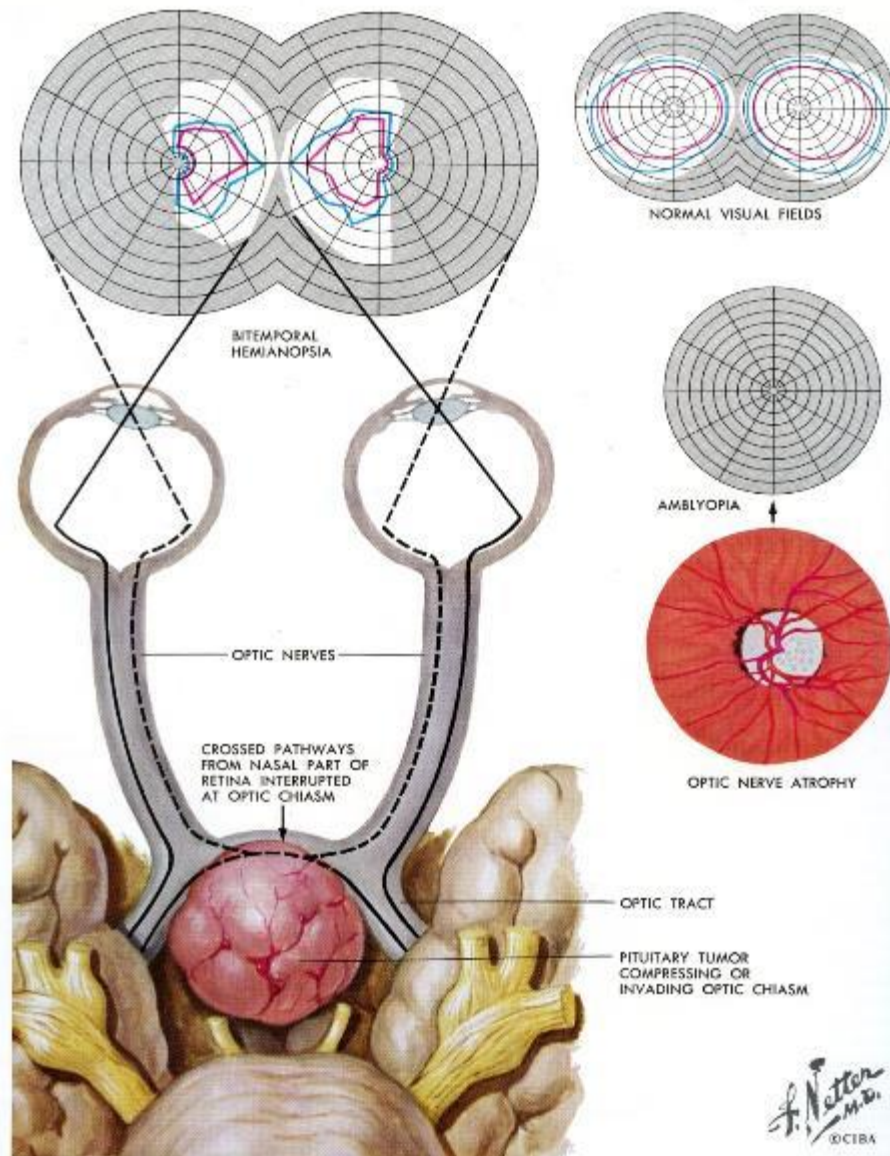
excess GH secretion: onset during childhood, puberty - **Gigantism**
onset after puberty - **Acromegaly**

impaired function: pituitary tumor

clinical symptoms: accelerated growth
insulin resistance
mild carbohydrate intolerance

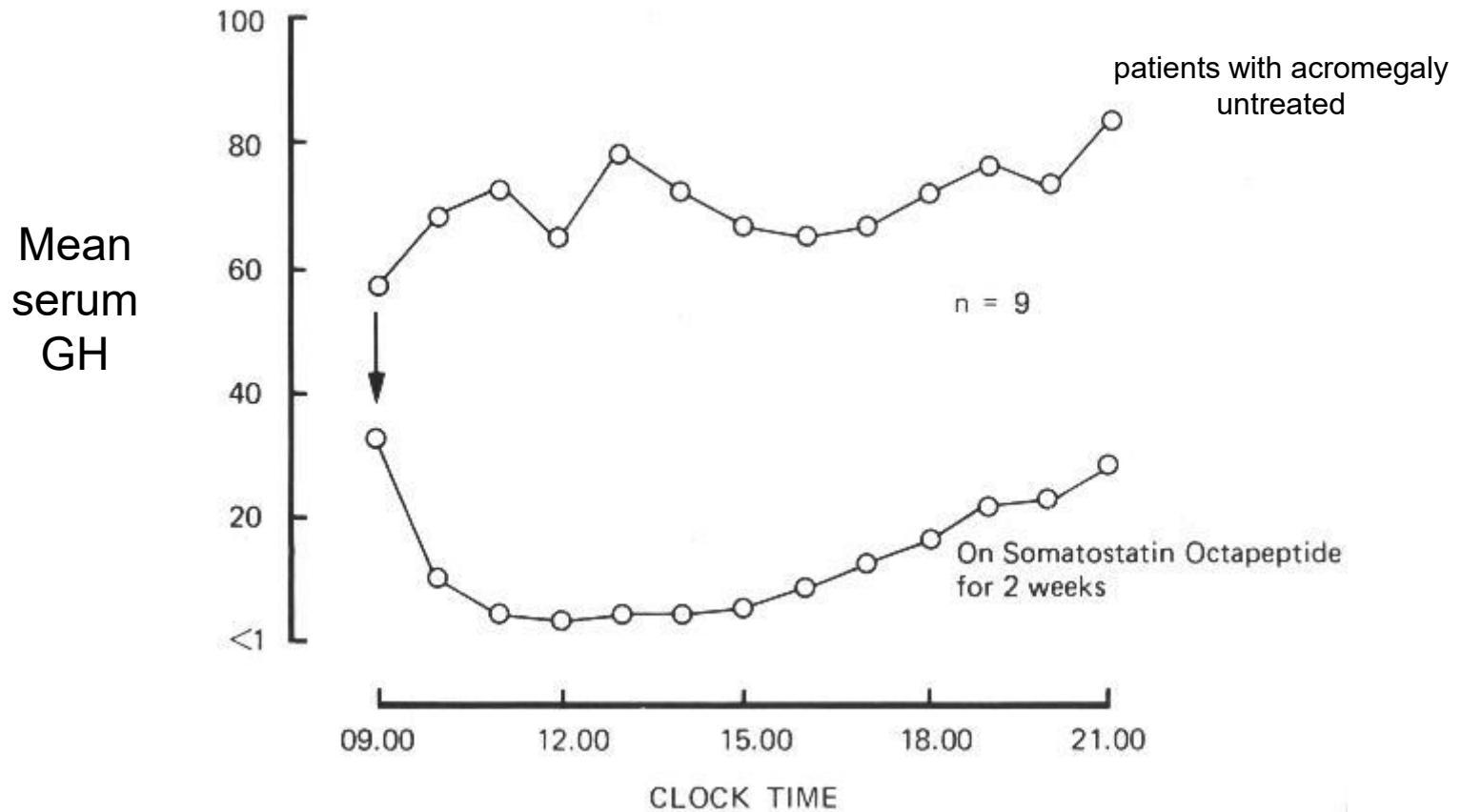
treatment: tumor removal (transphenoidal hypophysectomy)
somatostatin analogs (octreotide, lanreotide)

radiation (gamma knife radiosurgery)
dopamine agonists (bromocriptine, cabergoline, pergolide)
pegvisomant (GHR antagonist)



From Netter

Octreotide treatment



efficacy depends upon number of somatostatin receptors on tumor

GH Pathophysiology

Dwarfism

GH deficiency

impaired function:	hypothalamus - NO GHRH pituitary - NO GH or non-functional GH liver - NO IGF-I or IGF-II target tissue receptor defect: (number, binding, signaling)
clinical symptoms:	short stature delayed bone maturation mild obesity delayed puberty
treatment:	replacement therapy increased lean body mass, decreased adiposity children: increased growth velocity, pubescence, fertility

GH Pathophysiology

Dwarfism

GH deficiency

impaired function:

- hypothalamus - NO GHRH
- pituitary - NO GH or non-functional GH
- liver - NO IGF-I or IGF-II
- target tissue receptor defect: (number, binding, signaling)

Laron's syndrome: consanguinous parents (Asiatic Jewish population, isolated Ecuadoran population)

- apparent severe GH deficiency
- elevated circulating [GH]
- very low circulating [IGF-1]

Cause?

defective GH receptors

GH Pathophysiology

Dwarfism

GH deficiency

impaired function: hypothalamus - NO GHRH
 pituitary - NO GH or non-functional GH
 liver - NO IGF-I or IGF-II
 target tissue receptor defect: (number, binding, signaling)

Central African Forager population:

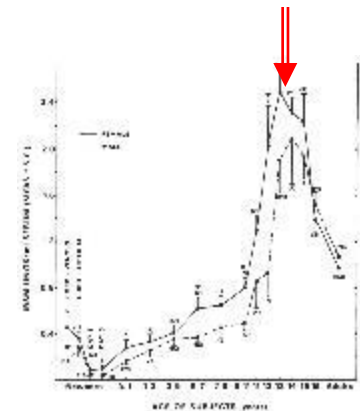
children: normal GHR

adults: normal GH, reduced IGF-1

Cause?

developmental regulation:

IGF-1 fails to increase at puberty



GH Pathophysiology

Dwarfism

GH deficiency

impaired function:

- hypothalamus - NO GHRH
- pituitary - NO GH or non-functional GH
- liver - NO IGF-I or IGF-II
- target tissue receptor defect: (number, binding, signaling)

Kwashiorkor: disease of children in some developing nations
severe protein/caloric deficiency
elevated GH

Cause?

GHR defects

Thank you.

Any questions?